

1 SchemeCANIFPPCT

This scheme is applicable to symmetric and asymmetric groups of prime orders.

1.1 Setup(n, m) $\rightarrow$ (mpk, msk)

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 $g_1 \leftarrow 1_{\mathbb{G}_1}$ 
 $g_2 \leftarrow 1_{\mathbb{G}_2}$ 
generate  $g_3 \in \mathbb{G}_1$ 
 $H_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$ 
 $H_2 : \mathbb{G}_T \rightarrow \mathbb{Z}_r$ 
 $H_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_r$ 
 $H_4 : \mathbb{G}_1 \rightarrow \mathbb{Z}_r$ 
generate  $r, s, t, \omega, t_1, t_2, t_3, t_4 \in \mathbb{Z}_r$  randomly
 $R \leftarrow g_1^r$ 
 $S \leftarrow g_2^s$ 
 $T \leftarrow g_1^t$ 
 $\Omega \leftarrow e(g_1, g_2)^{t_1 t_2 \omega}$ 
 $v_1 \leftarrow g_2^{t_1}$ 
 $v_2 \leftarrow g_2^{t_2}$ 
 $v_3 \leftarrow g_2^{t_3}$ 
 $v_4 \leftarrow g_2^{t_4}$ 
 $mpk \leftarrow (g_1, g_2, g_3, H_1, H_2, H_3, H_4, R, S, T, \Omega, v_1, v_2, v_3, v_4)$ 
 $msk \leftarrow (r, s, t, \omega, t_1, t_2, t_3, t_4)$ 
return ( $mpk, msk$ )

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1.2 KGen(ID_i, L) $\rightarrow$ (sk_{ID_i}, ek_{ID_i})

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generate  $k_i, x_i \in \mathbb{Z}_r$ 
 $z_i \leftarrow \frac{r - x_i}{s x_i}$ 
 $Z_i \leftarrow g_1^{z_i}$ 
 $sk_{ID_i} \leftarrow k_i$ 
 $ek_{ID_i} \leftarrow (x_i, Z_i)$ 
 $tag_i \leftarrow H_4(x_i \cdot Z_i)$ 
 $L \leftarrow L || ((ID_i, k_i, tag_i))$ 
return ( $sk_{ID_i}, ek_{ID_i}$ )

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1.3 Encryption($TPs_i, sk_{ID_i}, ek_{ID_i}, s, s_i$) $\rightarrow CT_{TP_i}$

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generate  $s_{1_i}, s_{2_i} \in \mathbb{Z}_r$  randomly
 $V_i \leftarrow H_2(\Omega^{s_i}), \forall i \in \{1, 2, \dots, n\}$ 
 $C_{0_i} \leftarrow (g_3 H_1(t_i || p_i))^{s_i}$ 
 $C_{1_i} \leftarrow v_1^{s_{1_i}}$ 
 $C_{2_i} \leftarrow v_2^{s_{1_i}}$ 
 $C_{3_i} \leftarrow v_2^{s_{1_i} - s_{2_i}}$ 
 $C_{4_i} \leftarrow v_2^{s_{1_i}}$ 

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Compute $a_0, a_1, a_2, \dots, a_n$ that satisfy $\forall x \in \mathbb{Z}_r$, we have $f(x) = \prod_{i=1}^n (x - V_i) =$

$$a_0 + \sum_{i=1}^n a_i x^i$$

generate $\alpha \in \mathbb{Z}_r$
 $C_1 \leftarrow g_1^\alpha$
 $C_2 \leftarrow Z_i^{x_i} + T^\alpha$
 $C_3 \leftarrow e(T, S)^\alpha$
 $C_4 \leftarrow H_3(C_{0_1} || C_{0_2} || \dots || C_{0_n} || C_{1_1} || C_{1_2} || \dots || C_{1_n} || \dots || C_{4_1} || C_{4_2} || \dots || C_{4_n} || a_0 || a_1 || \dots || a_{n-1} || C_1 || C_2 || C_3)$
 $C_5 \leftarrow sk_{ID_i} C_4 + x_i$
 $CT_{TP_s} \leftarrow (\vec{C}_0, \vec{C}_1, \vec{C}_2, \vec{C}_3, \vec{C}_4, C_1, C_2, C_3, C_4, C_5)$
return CT_{TP_i}

1.4 TokenGen(QTP_i, sk_{ID_i}) $\rightarrow token_i$

generate $r_{1_i}, r_{2_i} \in \mathbb{Z}_r$ randomly
 $T_{0_i} \leftarrow g_2^{r_{1_i} t_1 t_2 + r_{2_i} t_3 t_4}$
 $T_{1_i} \leftarrow g_1^{\omega t_2} (g_3 H_1(qt_i || qp_i))^{-r_{1_i} t_2}$
 $T_{2_i} \leftarrow g_1^{\omega t_1} (g_3 H_1(qt_i || qp_i))^{-r_{1_i} t_1}$
 $T_{3_i} \leftarrow (g_3 H_1(qt_i || qp_i))^{-r_{2_i} t_4}$
 $T_{4_i} \leftarrow (g_3 H_1(qt_i || qp_i))^{-r_{2_i} t_3}$
 $token \leftarrow (T_{0_i}, T_{1_i}, T_{2_i}, T_{3_i}, T_{4_i})$
return $token_i$

1.5 Query($CT_{TP_i}, token_i, s$) $\rightarrow y, y \in \{0, 1\}$

$V_i \leftarrow H_2(\Omega^{s_i}), \forall i \in \{1, 2, \dots, n\}$
 Compute $a_0, a_1, a_2, \dots, a_n$ that satisfy $\forall x \in \mathbb{Z}_r$, we have $f(x) = \prod_{i=1}^n (x - V_i) =$
 $a_0 + \sum_{i=1}^n a_i x^i$
 $V'_i \leftarrow H_2(e(C_{0_i}, T_{0_i})e(C_{1_i}, T_{1_i})e(C_{2_i}, T_{2_i})e(C_{3_i}, T_{3_i})e(C_{4_i}, T_{4_i}))$
return $f(x) = a_0 + \sum_{j=1}^n a_j V_i'^j = 0$

1.6 Trace(TP_{TP_i}, L) $\rightarrow identity$

$identity \leftarrow \perp$
for $element \in L$ **do**
 if $tag_i = element_2$ **then**
 $identity \leftarrow element$
 end if
end for
return $identity$